

MONISH KUMAR.K

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TECHNICAL SKILLS

Programming Language : C, Embedded C, Baremetal Programming, C++, DSA, QT(C++).
Microcontrollers : STM32F401CCU6 , PIC16F877A ,ESP32 and ESP8286.
Operating Systems : Linux (System Calls (File Operations,Memory Map)).
Tools : GCC, GDB, Make,MPLAB X IDE, Proteus, Kicad,Solidwork.
Scripting Language : Shell Script, Python.

INTERNSHIP EXPERIENCE

1.Mechimed Technologies Private Limited

Role: Product Developer Intern

Portable Finger-Based ECG System: Designed a system to measure ECG signals using finger electrodes, incorporating op-amp -based filters for noise reduction and signal quality enhancement.

Integrated ESP32 for real-time wireless data transmission, enabling seamless IoT-based health monitoring and analysis.

2. Epical Layouts Private Limited Company

Role: Product Developer Intern

Project: Invasive Glucometer DeviceThe system utilized blood sample test strips to measure glucose levels, with an analog front-end involving **amplification, filtering, and analog-to-digital conversion** for accurate readings.

PROJECTS

Project Name: Smart Wheelchair with AI Assistance (Collaboration with Purdue University)

Project Description: (FEB 2024 – Present)

Developed an AI-assisted smart wheelchair to enhance mobility for users with disabilities. Implemented voice-controlled navigation and real-time obstacle detection using ultrasonic sensors. Integrated IoT capabilities for remote monitoring and control. Designed an ergonomic and adaptive control system in collaboration with Purdue University to improve accessibility and ease of use.

Project Name: Portable IoT-Enabled Finger ECG Monitoring System

Project Description: (Nov 2024 – Dec 2024)

Designed a portable ECG system capable of acquiring signals from finger electrodes.Integrated ESP32 for wireless data transmission, enabling real-time monitoring and IoT-based healthcare analysis.

EDUCATION

Bachelor's degree in Electronics and Communication Engineering	CGPA-8.9
Chennai Institute of Technology, Chennai, Tamil Nadu	(2023-2027)